BrightLight Tutorials

Data Analytics

Exercise 1: SQL Fundamentals

Database: Employees_db

Table called: Employees

id	firstname	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. SELECT Statement

```
Select *
From ~~employees;~~ 'workspace'.'default'.'employees_table';
```

→ Return Results: It will return all the columns from the Employees table above.

2. SELECT DISTINCT Statement

```
SELECT DISTINCT department
FROM ~~employees~~; 'workspace'.'default'.'employees_table';
```

-- Return results:

department
IT
HR
Finance
Marketing

3. ORDER By Statement

```
SELECT
        first_name,
        last_name,
        Salary
FROM ~~employees~~ 'workspace'.'default'.'employees_table';
ORDER By Salary DESC;
```

--- Return results:

	first_name	last_name	Salary
1	Emily	Davis	62000
2	Mike	Johnson	60000
3	Robert	Wilson	59000
4	John	Doe	55000
5	Sarah	Brown	53000
6	Daniel	Clark	53000

	first_name	last_name	Salary
7	David	White	52000
8	Jessica	Moore	51000
9	Laura	Hall	5000
10	Jane	Smith	48000

4. LIMIT Statement

```
Select
        first_name,
        last_name,
        Salary
from 'workspace'.'default'.'employees_table'
Order by Salary desc
limit 5;
```

- Return Results:

	First_name	last_name	Salary
1	Emily	Davis	62000
2	Mike	Johnson	60000
3	Robert	Wilson	59000
4	John	Doe	55000
5	Sarah	Brown	53000

5. WHERE Statement

```
Select *
from 'workspace'.'default'.'employees_table
where department = 'IT';
```

- Return Results:

id	First_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6. AND Statement

```
Select *
from 'workspace'.'default'.'employees_table'
where department = "finance"
and Salary>58000;
```

- Return Results:

id	first_name	last_name	department	Salary	hire_date	City
3	Mike	Johnson	finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	finance	59000	2019-10-01	Houston

7. OR Statement

```
Select *
From 'workspace'.'default'.'employees_table'
where department = "HR"
or department = "Marketing";
```

- Return Results:

id	first_name	last_name	department	Salary	hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

8. NOT Statement

```
Select *
frome 'workspace'.default'.'employees_table'
where department <> 'IT';
```

- Return Results:

id	first_name	last_name	department	Salary	hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

9. IN Statement

```
Select *
From 'workspace'.default'.'employees_table'
where department in ('HR','IT')
or department = 'Finance';
```

- Return Results:

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

10. Combining Conditions

```
Select *
From 'workspace'.'default'.'employees_table'
where department = 'IT' and Salary > 50000
and City = 'New York';
```

-- Return Results:

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

11. Combining WHERE, AND, and ORDER BY

```
Select first_name, last_name, Salary, department
from 'workspace'.'default'.'employees_table'
Where department = 'finance' or department = 'Marketing'
and Salary > 52000
order by Salary desc;
```

-- Return Results

First_name	last_name	Salary	department
Mik	Johnson	60000	Finance
Robert	Wilson	59000	finance
Daniel	Clerk	53000	Marketing

12. Combining SELECT DISTINCT, WHERE, and IN

```
Select DISTINCT City
From 'workspace'.'default'.'employees_table'
where department not in ('IT', 'HR');
```

:- Return Results:

City
Chicago
Los Angeles
San Francisco
Houston

13. Combining WHERE, NOT, AND, and ORDER By ascending order.

```
Select *
From 'workspace'.'default'.'employees_table'
where department not in ('finace')
and Salary >50000
order by hire_date ASC;
```

Returns Results

id	first_name	last_name	department	salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
1	John	Doe	IT	55000	2018-06-15	New York
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

14. Combining WHERE, OR, IN and LIMIT

```
Select *
from 'workspace.default.employees_table'
where city = 'Chicago' or city = 'Los Angeles'
and department in ('IT', 'Marketing')
limit 3;
```

Return Results:

id	first_name	last_name	department	salary	hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago